

Problem-Solution Fit

Date	28 june 2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS]  - Small and marginal farmers - Large scale farmers - Agricultural consultants - Agri-researchers & academicians - Government & policy planners - Agri businesses & NGOs	6. CUSTOMER LIMITATIONS (e.g., Budget, Devices) [CL]  - High cost of soil testing & expert advice - Lack of knowledge about soil health & suitable crops - Limited access to real-time weather data - No data-driven decision making - Low digital literacy in rural areas - Limited access to advanced technology and tools	5. AVAILABLE SOLUTIONS (Pros & Cons) [AS]  - Traditional farming (experience based) - Low accuracy, high risk - Local agricultural experts - Costly, not always available - Mobile apps (basic information) - General info, not personalized - Govt. portals & schemes - Hard to understand, not integrated
2. PROBLEMS / PAINS + ITS FREQUENCY [PR]  - Choosing wrong crop for the soil (Very High) - Low crop yield & productivity (High) - Wastage of water, fertilizers & money (High) - Unpredictable weather & climate impact (High) - Lack of personalized guidance (High)	9. PROBLEM ROOT / CAUSE [RC]  - Lack of soil health analysis - No integration of soil + weather + crop data - Decisions based on guesswork or old experience - Non-availability of simple and reliable prediction tools - Information not in local language or farmer-friendly format	7. BEHAVIOR + ITS INTENSITY [BE]  - Farmers try different crops based on experience (High) - Depend on neighbors or local dealers for suggestions (High) - Hesitant to adopt new technologies (Medium to High) - Will switch if solution is accurate, easy & affordable (High)
3. TRIGGERS TO ACT [TR]  - Low yield / crop failure - Pest attacks & disease - Sudden weather changes - Increasing input costs - Need to improve income & productivity 4. EMOTIONS BEFORE / AFTER [EM]  Before: Worried, confused, stressed, unsure, taking risks After: Confident, relaxed, satisfied, informed, more productive	10. YOUR SOLUTION [SL]  OptiCrop – Smart Agricultural Production Optimization Engine - Soil, weather & crop data based smart recommendations - Machine Learning model for best crop prediction - Easy to use web/mobile platform - Real-time weather updates - Soil health improvement suggestions - Multi-language support for local farmers - Cost-effective and accessible solution	8. CHANNELS OF BEHAVIOR [CH]  ONLINE: - Website, Mobile App - Social Media (YouTube, Facebook, WhatsApp) - Agriculture Forums & Communities OFFLINE: - Krishi Melas, Workshops - Local Agri Centers, NGOs - Extension Officers, Cooperatives